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 Studierichting: Informatica
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$$\begin{aligned}
 & \int \frac{x - \sin x}{2 \sin^2 \frac{x}{2}} dx \\
 &= \int \frac{x - 2 \sin \frac{x}{2} \cdot \cos \frac{x}{2}}{2 \sin^2 \frac{x}{2}} dx \\
 &= \int \frac{x}{2 \sin^2 \frac{x}{2}} dx - \int \frac{\cos \frac{x}{2}}{\sin \frac{x}{2}} dx \\
 &= \frac{1}{2} \int x \cdot \frac{1}{\sin^2 \frac{x}{2}} dx - \int \frac{\cos \frac{x}{2}}{\sin \frac{x}{2}} dx \\
 & \left\{ \begin{array}{l} u = x \\ dv = \frac{1}{\sin^2 \frac{x}{2}} dx \end{array} \Rightarrow \left\{ \begin{array}{l} du = 1 dx \\ v = * \end{array} \right. \right. \\
 & * \Rightarrow t = \frac{x}{2} \Rightarrow dt = \frac{1}{2} dx \Rightarrow 2dt = dx \\
 & \Rightarrow \frac{1}{2} \int \frac{1}{\sin^2 t} \cdot 2dt = \int \frac{1}{\sin^2 t} dt = -\cot t = -\cot \frac{x}{2} \\
 &= \left[x \cdot \left(-\cot \frac{x}{2} \right) - \int -\cot \frac{x}{2} \cdot 1 dx \right] - \left[\int \frac{\cos \frac{x}{2}}{\sin \frac{x}{2}} dx \right] \\
 &= -x \cot \frac{x}{2} + \int \cot \frac{x}{2} dx - \int \cot \frac{x}{2} dx \\
 &= -x \cot \frac{x}{2} + C
 \end{aligned}$$

References